

MORINGA-ANGANI BI HACKATHON CASE STUDY

19th September 2026

ANGANI AIR: TURNING LOYALTY DATA INTO CUSTOMER GROWTH

► The Business Situation

Angani Air (AA) is a growing African airline competing in an increasingly competitive travel market. To strengthen customer loyalty and encourage repeat travel, AA launched a **Loyalty Program Promotion between February and April 2018**.

The campaign generated a significant number of new loyalty-program signups. However, senior management now faces a more important question:

“Did the 2018 Loyalty Program Promotion create valuable customer behaviour, and how should Angani Air improve its loyalty strategy?”

Management wants to understand:

- Whether the campaign **changed customer behaviour**
- **Where AA should focus its future loyalty investment.**

The dataset contains **customer**, **loyalty-program** and **flight information** that can be used to investigate this question.

► The Business Data

You can download the dataset* for this task from <https://smplu.link/BIHACKATHON>

► Your Task

You have been hired as a **Business Intelligence & Customer Strategy team** by Angani Air. Your team has been given the company's loyalty and flight data and has been asked to build a **management decision-support solution**. You are **not being asked simply to build a dashboard**.

Your job is to:

**Understand Business → Define Decisions → Identify Questions →
Select KPIs → Storyboard → Build Dashboards**

* Dataset Note: This case study uses an adapted version of a publicly available airline dataset. Company names and contextual elements have been modified for the purposes of this hackathon

► The Core Business Question

"Did the 2018 Loyalty Program Promotion create valuable customer behaviour, and how should Angani Air improve its loyalty strategy?"

Your analysis should ultimately allow management to answer:

- **Who should we target**
- **What should we offer them**
- **Where should we invest our loyalty budget?**

► Hackathon Challenge

Teams must develop a **Business Intelligence solution** that evaluates the performance of AA's loyalty-program promotion and **identifies opportunities** to improve customer engagement.

You should investigate the relationship between:

**Campaign → Loyalty Membership → Customer Characteristics →
Flight Behaviour → Customer Value.**

The final solution should move beyond reporting what happened and **provide credible insights and actionable business recommendations for management.**

The **5 key criteria** used for the final judgement are outlined in the 'Final Presentation' section.

► Required Tools

1. Business Intelligence / Visualisation

Participants should use [Microsoft Power BI](#).

2. Data Cleaning, Transformation and Analysis

Participants may use:

- **Excel, Python and SQL** for data cleaning, exploration and accuracy checks
- **The final BI Solution (Dashboard) must be created in Microsoft Power BI**

3. AI Use

- AI tools such as ChatGPT, Copilot and Gemini are permitted. However, teams are responsible for understanding, validating and being able to explain any AI-generated output used in their solution.

► Deliverables

- Each team is expected to submit an Interactive BI Solution (dashboards) using **Power BI**
- **Maximum: 5 reports pages** → Each tab in a Power BI file is called a report page.
NB: More pages ≠ more marks. Focus on quality, not quantity.
- The solution should demonstrate:
 - Appropriate data preparation
 - Appropriate data modelling
 - Meaningful KPIs
 - Analytical depth
 - Effective visualisation
 - Interactivity
 - Clear business Story Telling
 - Actionable insights
- Your final PBIX Report should be emailed to: **bidasanalytics@gmail.com**

► Final Presentation

- Each team will present its solution to the other groups as if they were the **AA Executive Committee**.
- Each group will appoint a leader, who will share their number and receive an **online scoring** form link via **WhatsApp** to enter the scores.
- The groups will judge each other based on the **5 criterion below**. Rate each criterion from **1 (Poor) to 10 (Excellent)**.

Criterion	What is being Judged	Maximum Score
Business Understanding	The dashboard addresses a real problem and shows clear understanding of the domain, users, and business context	10
Dashboard Design & Interactivity	The dashboard is visually clear, well laid out, and interactive (filters, drill-downs, hover states) in a way that aids understanding.	10
Story Telling & Narrative Flow	The dashboard guides the viewer through a logical narrative from problem → insight → conclusion.	10
Decision-Focused Insights	The dashboard surfaces meaningful, non-obvious insights and clearly supports a decision or action – a viewer knows what to do next.	10
Final Presentation	The group presented the dashboard clearly, confidently, and within the time limit.	10

- The group with the **highest overall score** will be **declared the winner**. In the event of a tie, the judges will determine the winner.

Maximum presentation time: 10 minutes per group

- The presentation should focus on:

What happened → Why it matters → What management should do next

- NB: Teams should **not simply give a page-by-page walkthrough of their dashboard**.

NO PRESSURE...

BUT YOUR DASHBOARD BETTER EAT!! 😊